THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
041 BASIC MATHEMATICS
MARKING SCHEME NOVEMBER 2024

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1. (a) $1\,109^2 = 1\,229\,881$..... 01 mark
$\sqrt{841} = \pm 29$..... 01 mark
The least or lowest sum $= 1\,229\,881 - 29 = 1\,229\,852$..... 01 mark

$1\,229\,852 =$ One million, two hundred twenty nine thousand, eight hundred fifty two....02 mark

(b) $0.54545\ldots = 0.\overline{54}$

Let $0.\overline{54}\ be\ y$

$y = 0.\overline{54}$.....equation 1.....01 mark

$y \times 100 = 0.\overline{5454} \times 100 = 54.54$

$100y = 54.54$ .....equation2.....01 mark

Subtract equation 1 from equation 2

$$-\left|\begin{matrix} 100y = 54.54 \\ y = 0.54 \end{matrix}\right.$$

$99y = 54$.....01 mark

$\frac{99y}{99} = \frac{54}{99}\ \ y = \frac{54\div 9}{99\div 9} = \frac{6}{11}$.....01 mark.

Fraction for girls $= 1 - \frac{6}{11} = \frac{5}{11}$.....01 mark

2. (a) i. $0.080579 \approx 0.081$ ii. $0.080579 \approx 0.0806$ ..... 05 marks

(b)

	km	dam	cm
	5	320	409
×			8
	65	63	272

....05 marks

3. (a) Area of a circle $(A) = \pi r^2$.....$00\frac{1}{2}$ mark
$= 3.14 \times 5cm \times 5\ cm$
$= 78.5\ cm^2$.....01 mark
Area of a square$= \frac{diagonal_1 \times diagonal_2}{2} = \frac{d_1 \times d_2}{2}$.....$00\frac{1}{2}$ mark..
$= \frac{10\ cm \times 10\ cm}{2} = 50\ cm^2$.....01 mark

Alternatively

Area of a square $= Area\ of\ \Delta AOD\ \times 4$..... $00\frac{1}{2}$ mark

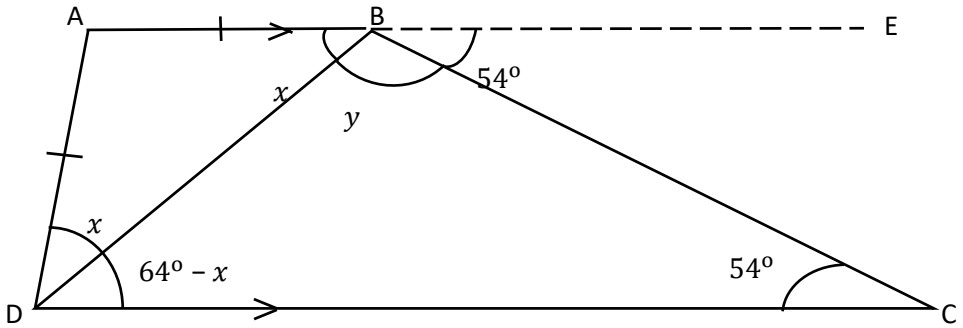
$= \frac{1}{2} \times 5\ cm\ \times 5\ cm \times 4$ ..... $00\frac{1}{2}$ mark

$= 50\ cm^{\ 2}$.....$000\frac{1}{2}$ mark

Area of the shaded region $= 78.5\ cm^2 - 50\ cm^2 = 28\ .5\ cm^2$.....01 mark

The area of the shaded region is about $28\ cm^2$(8 is an even digit)..... .....01 mark

(b) Extend line $\overline{AB}\ to\ E$



$\angle DCB = \angle EBC\ (\overline{AB} \parallel \overline{DC}\ alternate\ interior\ angles)$.....$00\frac{1}{2}$ mark

$\angle ADB = \angle ABD(Base\ angles\ of\ isoscless\ triangle)$ ..... $00\frac{1}{2}$ mark

Let $\angle DBC\ be\ y\ and \angle ABD\ be\ x$

$x + y + 54^\circ = 180^\circ$ (Straight-line angle)

$x + y = 180^\circ - 54^\circ$

$x + y = 126^\circ$ .....equation 1 ..... $00\frac{1}{2}$ mark

$\angle BDC = 64^\circ - x$

m$64^\circ - x + 54^\circ + y = 180^\circ$

$y - x = 180^\circ - (64^\circ + 54^\circ)$

$y - x = 62^\circ$.....equation 2..... $00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$$+\left|\begin{matrix} y + x = 126^\circ \\ y - x = 62^\circ \end{matrix}\right.$$

$2y = 188^\circ$ ..... $00\frac{1}{2}$ mark

$\frac{2y}{2} = \frac{188^\circ}{2}$

$y = \angle DBC = 94^\circ$.....$00\frac{1}{2}$ mark

Alternatively

Extend line $\overline{AB}\ to\ E$

$\angle ADB = \angle ABD$ (Base angles of isoscless triangle ABD).....01 mark

$\angle ABD = \angle CDB$(Alternative interior angles $\overline{AB} \parallel \overline{CD}$).....01 mark

$\angle DCB = \angle EBC$ (Alternate interior angles $\overline{AB} \parallel \overline{CD}$)..... $00\frac{1}{2}$ mark

$\angle ABD = \frac{1}{2}\angle ADC = \frac{1}{2} \times 64^\circ = 32^\circ$ .....01 mark

$\angle ABD + \angle DBC + \angle EBC = 180^\circ$ (Straight line angle)..... . $00\frac{1}{2}$ mark

$$\angle DBC = 180^\circ - (32^\circ + 54^\circ)$$
$$= 180^\circ - 86^\circ$$
$$= 94^\circ$$ .....01 mark

4. (a) $\left(\frac{9-r}{2}\right) = 14$

$$9 - r = 28$$
$$r = 9 - 28$$
$$r = -19$$..... $02\frac{1}{2}$ marks

$$2(r - 9) = 2(-19 - 9)$$
$$= 2(-28)$$
$$= -56$$..... $02\frac{1}{2}$ marks

(b) $15x - y = 9$
$7x - 2y = 12$

$2 \mid 15x - y = 9$
$1 \mid 7x - 2y = 12$

$30x - 2y = 18$
$7x - 2y = 12$ .....$00\frac{1}{2}$ mark

$- \mid 30x - 2y = 18$
$\quad\;\; 7x - 2y = 12$

$23x = 6$ ..... 01 mark

$\frac{23x}{23} = \frac{6}{23}$

$x = \frac{6}{23}$ ..... 01 mark

$15x - y = 9$
$7x - 2y = 12$

$7 \mid 15x - y = 9$
$15 \mid 7x - 2y = 12$

$105x - 7y = 63$
$105x - 30y = 180$.....$00\frac{1}{2}$ mark

$- \mid 105x - 7y = 63$
$\quad\;\; 105x - 30y = 180$

$23y = -117$..... 01 mark

$\frac{23y}{23} = \frac{-117}{23}$

$y = \frac{-117}{23}$ ..... 01 mark

5.(a) Total parts $= 2 + 4 + 5 + 9 = 20$ parts.....01 mark

Seypalm's share $= \frac{2}{20} \times 80\,000 = 8\,000$ shillings .....01 mark

Joseph's share $= \frac{4}{20} \times 80\,000 = 16\,000$ shillings ..... 01 mark

Juma's share $= \frac{5}{20} \times 80\,000 = 20\,000$ shillings.....01 mark

Yasinta's share $= \frac{9}{20} \times 80\,000 = 36\,000$ shillings.....01 mark

(b) $p = \frac{I}{rt} = \frac{108\,000}{0.05 \times 5}$.....01 mark

$= \frac{108\,000}{0.25}$.....01 mark

$= \frac{108\,000 \times 100}{0.25 \times 100}$.....01 mark

$= \frac{10\,800\,000}{25}$.....01 mark

$= 432\,000$ Shillings

Therefore 432 000 shillings is the principal that would earn 108 000 shillings at the rate of 5% per annum in 5 years.....01 mark

6.(a) i. Solve for y from $2x - 3y + 7 = 0$

$3y = 2x + 7$.....01 mark

$\frac{3y}{3} = \frac{2x+7}{3}$.....01 mark.

$y = \frac{2x}{3} + \frac{7}{3}$.....01 mark

ii. Gradient or slope $= m = \frac{2}{3}$.....01 mark

iii. $y-$ Intercept $= c = \frac{7}{3}$.....01 mark

(b) $m = \frac{y_2 - y_1}{x_2 - x_1}$.....01 mark

$2 = \frac{4-2}{4-x}$.....01 mark

$2(4 - x) = 2$

$8 - 2x = 2$.....01 mark

$8 - 2 = 2x$

$6 = 2x$.....01 mark

$\frac{2x}{2} = \frac{6}{2}$

$x = 3$.....01 mark

7. (a) 4 kg $=$4 000 g.....01 mark

4 kg 30 g $= 4\,000g + 30\ g = 4\,030\ g$.....01 mark

20 kg $= 20\,000\ g$.....01 mark

20 kg 150 g $= 20\,000\ g + 150g = 20\,150\ g$.....01 mark

4 030 g $=$ 1 hour

20 150 g $= y\ hours$

$y = \frac{1\ hour \times 20\,150\ g}{4\,030\ g}$

$= 5\ hours$.....01 mark

(b) $-5 < 3x + 1 \leq 13$

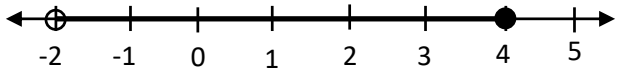
$-5 - 1 < 3x + 1 - 1 \leq 13 - 1$..... $00\frac{1}{2}$ mark

$-6 < 3x \leq 12$ .....01 mark

$\frac{-6}{3} < \frac{3x}{3} \leq \frac{12}{3}$.....$00\frac{1}{2}$ mark

$-2 < x \leq 4$ .....01 mark

$-2 < x \leq 4$



.....02 marks

8. (a) 37.5% $= \frac{3}{8}$.....$00\frac{1}{2}$ mark

English $\frac{3}{8} \times 240 = 90$ students.....01 mark

Art $\frac{1}{6} \times 240 = 40$ students..... $00\frac{1}{2}$ mark

Total $= 90 + 40 = 130$ students.....01 mark

Hence $240 - 130 = 110$ students prefer Mathematics.....01 mark

$110 - 90 = 20$

Therefore 20 more students prefer Mathematics tha English.....01 mark.

Alternative

Fraction for students prefer Mathematics $= 1 - \left(\frac{3}{8} + \frac{1}{6}\right)$

$= 1 - \left(\frac{9+4}{24}\right)$

$= 1 - \frac{13}{24}$.....$00\frac{1}{2}$ mark

$= \frac{24-13}{24}$

$= \frac{11}{24}$.....01 mark

Students who prefer more Mathematics than English $= \left(\frac{11}{24} - \frac{3}{8}\right) \times 240$.....01 mark

$= \left(\frac{11-9}{24}\right) \times 240$..... $00\frac{1}{2}$ mark

$= \frac{2}{24} \times 240$.....01 mark

$= 20$ students.....01 mark

(b) i. Prime numbers are $\{31 , 41 , 61 , 71 , 101\}$.....$02\frac{1}{2}$ marks

ii. Two equivalent fractions are $\frac{1}{7}\ and \frac{13}{91}\ or \frac{7}{1}\ and \frac{91}{13}$ ..... $02\frac{1}{2}$ marks

9.(a) 40% of 3 days $= \frac{40}{100} \times 3 \times 24 \times 60$ minutes .....01 mark

$= 1\,728$ minutes .....01 mark

80% of 1 128 minutes $= 902.4$ minutes.....01 mark

Since 1 728 minutes > 902.4 minutes.....01 mark

Therefore 40% 0f 3 days is longer than 80% 0f 1128 minutes.....01 mark

(b) Area (A) = length (l) × width(w).....01 mark

$A = lw\ , l = \frac{A}{w} = \frac{50\ m^2}{2.5\ m} = 20\ m$ .....01 mark

Perimeter (p)$= 2l + 2w$.....01 mark

$= 2 \times 20\ m + 2 \times 2.5\ m$

$= 40\ m + 5\ m$.....01 mark

$= 45\ m$.....01 mark

10. (a) i. $(n-2) \times 180^{\circ} = 1\ 080^{\circ}$ .....$00\frac{1}{2}$ mark

$(n-2) \times \frac{180^{\circ}}{180^{\circ}} = \frac{1\ 080^{\circ}}{180^{\circ}}$

$n - 2 = 6$..... $00\frac{1}{2}$ mark

$n = 6 + 2$

$= 8$.....01 mark

Therefore, the regular polygon is an Octagon.....01 mark

ii. $8 - 8 \times \frac{2\frac{1}{5} - 2\frac{1}{7}}{2 - \frac{1}{6 - \frac{1}{6}}} = 8 - 8 \times \frac{\frac{11}{5} - \frac{9}{7}}{2 - \frac{1}{\frac{36-1}{6}}}$

$= 8 - 8 \times \frac{\frac{11}{5} - \frac{9}{7}}{2 - \frac{1}{\frac{35}{6}}}$

$= 8 - 8 \times \frac{77-45}{\frac{35}{2 - \frac{6}{35}}}$.....$00\frac{1}{2}$ mark .

$= 8 - 8 \times \frac{32}{\frac{35}{\frac{70-6}{35}}}$

$= 8 - 8 \times \frac{32}{35} \times \frac{35}{64}$.....$00\frac{1}{2}$ mark

$= 8 - 8 \times \frac{32}{64}$.....$00\frac{1}{2}$ mark

$= 8 - 8 \times \frac{1}{2}$.....$00\frac{1}{2}$ mark

$= 8 - 4$..... $00\frac{1}{2}$ mark

$= 4$..... $00\frac{1}{2}$ mark

(b) $18 = 2 \times 3 \times 3$, $22 = 2 \times 11$, $15 = 3 \times 5$.....01 mark

$GCF = 1$.....01 mark

$LCM = 2 \times 3 \times 3 \times 5 \times 11 = 990$.....01 mark

Sum of GCF and LCM $1 + 990 = 991$.....01 mark